Integer programming and branch and bound



Optimization · University · Lecture 2

In this lecture

- Model with integers.
- Use relaxations as bounds.
- Run branch and bound.

Remember

- Never just round.
- Relaxations bound.
- Prune by bound.

Grow the tree

1. Warm-up (5 min): Round it?
2. Relax (10 min): The LP.
3. Branch (25 min): The tree.
4. Compare (15 min): Rounding.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. How many nodes did you prune?
2. How far was rounding from optimal?
3. What does the gap tell a manager?

Exit ticket

- Name one decision in your institution that needs integer variables.